

Error in gradient calculation in qGAN using Gradient()

<https://github.com/qiskit-community/qiskit-machine-learning/issues/70>

ROW 106

Error in gradient calculation in qGAN using Gradient() #70

[New issue](#)[Closed](#)

sychang42 opened on Apr 26, 2021

Information

- Qiskit Machine Learning version: 0.1.0
- Python version: 3.8.8
- Operating system: Windows 10

What is the current behavior?

I was trying to apply "Phase shift rule" for gradient calculation in quantum GAN using the tutorial : https://qiskit.org/documentation/tutorials/machine_learning/04_qgans_for_loading_random_distributions.html.

To do so, I replaced the line :

```
qgan.set_generator(generator_circuit=g_circuit, generator_init_params=init_params)
```

by

```
qgan.set_generator(generator_circuit=g_circuit, generator_init_params=init_params, generator_gradient =
```

with `qiskit.aqua.operators.gradients.Gradient` .

However, while running

```
result = qgan.run(quantum_instance)
```

with `reps = 2` (2 blocks in the quantum generator), the code shows me an error message below :

```
ValueError                                Traceback (most recent call last)
~\Anaconda3\lib\site-packages\qiskit\aq\components\neural_networks\quantum_generator.py in loss(self, x, weights)
    382         # pylint: disable=no-member
--> 383         loss = (-1) * np.dot(np.log(x).transpose(), weights)
    384     except Exception: # pylint: disable=broad-exception
<__array_function__ internals> in dot(*args, **kwargs)
ValueError: shapes (4,1) and (6,4) not aligned: 1 (dim 1) != 6 (dim 0)
```

Please also find the full error message below

```
ValueError                                Traceback (most recent call last)
~\Anaconda3\lib\site-packages\qiskit\aq\components\neural_networks\quantum_generator.py in loss(self, x, weights)
    382         # pylint: disable=no-member
--> 383         loss = (-1) * np.dot(np.log(x).transpose(), weights)
    384     except Exception: # pylint: disable=broad-exception
<__array_function__ internals> in dot(*args, **kwargs)
ValueError: shapes (4,1) and (6,4) not aligned: 1 (dim 1) != 6 (dim 0)

During handling of the above exception, another exception occurred:

ValueError                                Traceback (most recent call last)
<ipython-input-13-009eadbfa5b5> in <module>
      1 # Run qGAN
----> 2 result = qgan.run(quantum_instance)

~\Anaconda3\lib\site-packages\qiskit\aq\algorithms\quantum_algorithm.py in run(self, quantum_instance, **kwargs)
    69         self.quantum_instance = quantum_instance
    70
--> 71         return self._run()
    72
    73     @abstractmethod

~\Anaconda3\lib\site-packages\qiskit\aq\algorithms\distribution_learners\qgan.py in _run(self)
    365         'Set backend either to statevector_simulator, qasm_simulator'
    366         'or actual quantum hardware')
--> 367         self.train()
    368
    369         return self._ret

~\Anaconda3\lib\site-packages\qiskit\aq\algorithms\distribution_learners\qgan.py in train(self)
    325         # 2. Train Generator
    326         self.generator.discriminator = self._discriminator
--> 327         ret_g = self.generator.train(self.quantum_instance, shots=self.batch_size)
    328         g_loss_min = ret_g['loss']
    329
```

Assignees

Zoufalac

Labels

priority: low **type: bug**

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants

3

Zot

```

~\Anaconda3\lib\site-packages\qiskit\aquas\components\neural_networks\quantum_generator.py in train(self, quantum_instance, shots)
494
495     objective = self._get_objective_function(quantum_instance, self._discriminator)
--> 496     self._bound_parameters, loss, _ = self._optimizer.optimize(
497         num_vars=len(self._bound_parameters),
498         objective_function=objective,

~\Anaconda3\lib\site-packages\qiskit\aquas\components\optimizers\adam_amsgrad.py in optimize(self, num_vars, objective_function, gradient_function, variable_bounds, initial_point)
241         (objective_function, self._eps))
242
243     point, value, nfev = self.minimize(objective_function, initial_point, gradient_function)
--> 244     return point, value, nfev

~\Anaconda3\lib\site-packages\qiskit\aquas\components\optimizers\adam_amsgrad.py in minimize(self, objective_function, initial_point, gradient_function)
180         A tuple of (optimal parameters, optimal value, number of iterations).
181         """
--> 182         derivative = gradient_function(initial_point)
183         self._t = 0
184         self._m = np.zeros(np.shape(derivative))

~\Anaconda3\lib\site-packages\qiskit\aquas\components\neural_networks\quantum_generator.py in gradient_function(current_point)
450         value_dict = {free_params[i]: current_point[i] for i in range(len(free_params))}
451         analytical_gradients = np.array(grad_object.assign_parameters(value_dict).eval())
--> 452         loss_gradients = self.loss(prediction_generated, analytical_gradients).real
453         return loss_gradients
454

~\Anaconda3\lib\site-packages\qiskit\aquas\components\neural_networks\quantum_generator.py in loss(self, x, weights)
383         loss = (-1) * np.dot(np.log(x).transpose(), weights)
384         except Exception: # pylint: disable=broad-except
--> 385             loss = (-1) * np.dot(np.log(x), weights)
386         return loss.flatten()
387

<__array_function__ internals> in dot(*args, **kwargs)

ValueError: shapes (1,4) and (6,4) not aligned: 4 (dim 1) != 6 (dim 0)

```

Steps to reproduce the problem

Please also find the code [here](#).

In the tutorial [04_qgans_for_loading_random_distributions.ipynb](#),

1. Import gradient object :

```
from qiskit.opflow import Gradient
```

2. Replace the line :

```
qgan.set_generator(generator_circuit=g_circuit, generator_init_params=init_params)
```

by

```
qgan.set_generator(generator_circuit=g_circuit, generator_init_params=init_params, generator_gradient =
```

3. Replace the line :

```
var_form = TwoLocal(int(np.sum(num_qubits)), 'ry', 'cz', entanglement=entangler_map, reps=1)
```

by

```
var_form = TwoLocal(int(np.sum(num_qubits)), 'ry', 'cz', entanglement=entangler_map, reps=2)
```

4. Add two more values for `init_params` (array of size 6)

5. Run the quantum GAN.

What is the expected behavior?

By the chain rule, the gradient of the generator is calculated as :



(c.f. Appendix B in [Supplementary Information: Quantum Generative Adversarial Networks for Learning and Loading Random Distributions](#))

As we are summing over [\[1\]](#), the matrix multiplication performed in

```
loss_gradients = self.loss(prediction_generated, analytical_gradients).real
```



(line 411 in [quantum_machine_learning.algorithms.distribution_learners.qgan.quantum_generator](#))

should take the form :

[\[2\]](#)

where

[\[2\]](#),

However,

```
analytical_gradients = np.array(grad_object.assign_parameters(value_dict).eval())
```



in

```
def _convert_to_gradient_function(self, gradient_object, quantum_instance, discriminator)
```



currently returns:

(line 410 in [quantum_machine_learning.algorithms.distribution_learners.qgan.quantum_generator](#))

[\[2\]](#),

so the code returns an error for matrix multiplication.

Suggested solutions

I think the analytical gradients should be transposed :

```
analytical_gradients = np.array(grad_object.assign_parameters(value_dict).eval())
```



to

```
analytical_gradients = np.transpose(np.array(grad_object.assign_parameters(value_dict).eval()))
```



[woodsp-ibm](#) assigned [Zoufalc](#) on Apr 26, 2021



[adekusal-drl](#) added **priority: low** **type: bug** on Apr 26, 2021



[Zoufalc](#) mentioned this on Apr 27, 2021

[Fix qGAN gradient behavior #77](#)

[adekusal-drl](#) on May 5, 2021

Collaborator ...

Closed in [#77](#)



[adekusal-drl](#) closed this as **completed** on May 5, 2021

Fix qGAN gradient behavior #77

<> Code

Merged adekuser-drl merged 16 commits into `qiskit-community:main` from `Zoufalc:qgan_grad_fix` on May 5, 2021

Conversation 5 Commits 16 Checks 0 Files changed 3 +21 -5



Zoufalc commented on Apr 27, 2021

Contributor

Summary

Address Issue #70 by fixing the dimensions of the gradients & include an additional gradient test for the statevector backend.

Details and comments



fix qgan grads

cd61c94

Zoufalc requested review from adekuser-drl, manaelmarques, pbark, stefan-woerner and woodsp-ibm as code owners 4 years ago



CLAassistant commented on Apr 27, 2021 • edited

...

CLA signed

All committers have signed the CLA.



include reno

b0671f0



woodsp-ibm reviewed on Apr 27, 2021

View reviewed changes

test/algorithms/distribution_learners/qgan/test_qgan.py Outdated

Show resolved

woodsp-ibm added the Changelog: Bugfix label on Apr 27, 2021

Zoufalc and others added 10 commits 4 years ago

update test for ddt

e16b713

Merge branch 'main' into qgan_grad_fix

Verified

9db4efa

fix lint

a571fc3

Merge branch 'qgan_grad_fix' of https://github.com/Zoufalc/qiskit-mac...

18e2445

fix test

98c25de

fix lint

76dc974

fix lint

5c6cfa0

delete test file

1bd555a

fix lint again

d5d0da9

Merge branch 'main' into qgan_grad_fix

Verified

e6dc741

Reviewers

woodsp-ibm



adekuser-drl



manaelmarques



pbark



stefan-woerner



Assignees

No one assigned

Labels

Changelog: Bugfix

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

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5 participants



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